

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1922

Lincoln 230 kV
March 21, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1922. The Interconnection Customer proposes a 241.1 MW natural gas combined cycle facility interconnecting at the Lincoln 230 kV substation in the Illinois area. The request was received on November 01, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 593 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,065,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

- Facility: Lincoln to Elwood 230kV Line
Contingency: Blythe to South Boston 230kV
Loading: 109.0% (Rating: 612 MVA)
DFAX: 38.1%
Pre-project loading: 94.0%
- Facility: S Boston to Blythe 230kV
Contingency: Lincoln Substation to Elwood 230kV
Loading: 103.5% (Rating: 714 MVA)
DFAX: 21.6%
Pre-project loading: 96.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lincoln to Elwood 230kV	Reconductor line	\$11,478,000	21% (\$2,412,000)
S Boston to Blythe 230kV	Reconductor line	\$13,446,000	12% (\$1,653,000)
	Total allocated to project		\$4,065,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.